STOCK 1 DRILLED I-BEAM · REV v2 · FINAL DESIGN

EMCH 308 Team 6 · AA I 3 × 2.03, 6061-T6 · 8.20 in long · 3-point bend, 8.00 in span

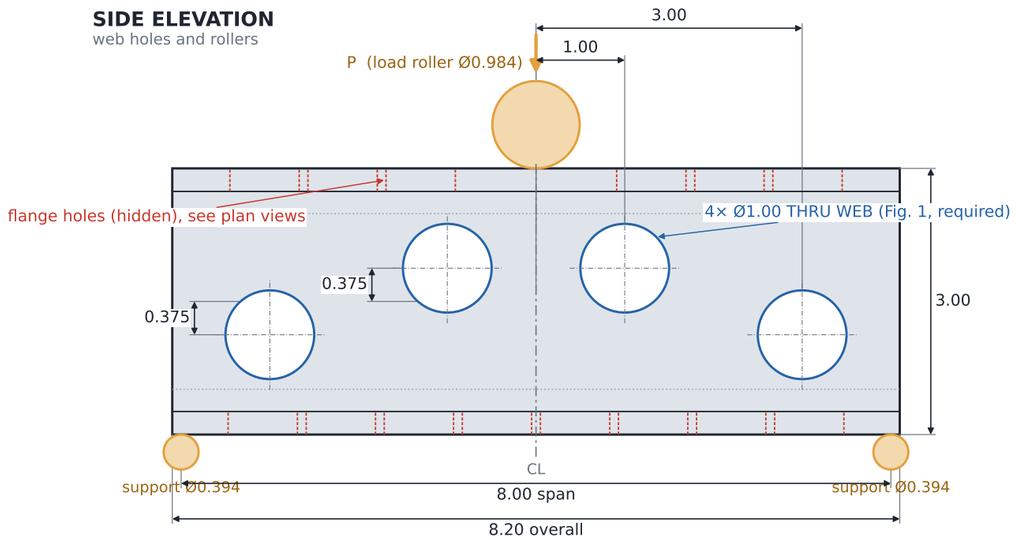
Dimensions in inches from mid-span (CL); symmetric about CL and the web centreline.

WEIGHT PASS
0.993 lb
limit ≤ 1.00 lb

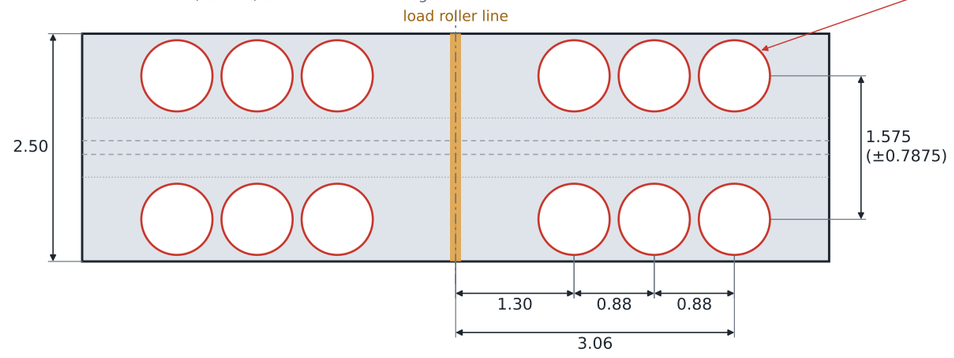
FACTOR OF SAFETY PASS
1.625
required ≥ 1.25

COST PASS
$143.47
limit ≤ $600

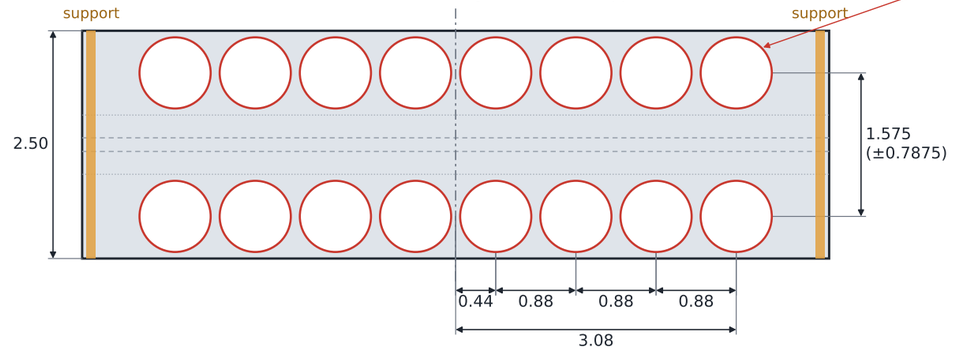
SIDE ELEVATION
web holes and rollers



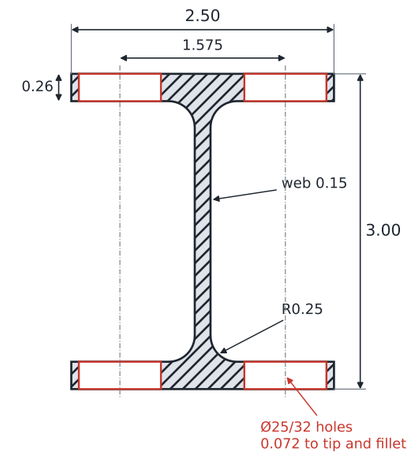
TOP FLANGE · PLAN
12 holes · stations ±1.30, ±2.18, ±3.06 · first hole edge 0.91 from the load roller



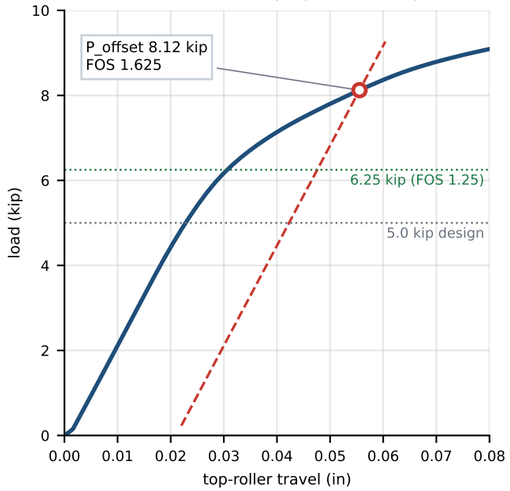
BOTTOM FLANGE · PLAN
16 holes · stations ±0.44, ±1.32, ±2.20, ±3.08 · last hole edge 0.53 from a support



SECTION A–A
through a flange-hole station



FOS · ABAQUS LOAD–DISPLACEMENT
0.02 in offset method · C3D10 mesh, 65,942 elements, 0 distorted



REQUIREMENTS CHECK

Requirement	Limit	Final design	
Weight	≤ 1.00 lb	0.993 lb (CAD = curved-element mesh)	PASS
Factor of safety	≥ 1.25	1.625 (P_offset 8,123 lbf)	PASS
Cost	≤ $600	$143.47	PASS
Span / length / height	fixed	8.00 / 8.20 / 3.00 in	PASS
Fig. 1 holes	4 × Ø1.00	kept, unchanged	PASS

COST · course cost sheet

Material, Stock 1 (8.4 in)	$55.02
Saw to 8.20 (1 cut)	$18.34
Drill, 32 holes	$70.11
Total	**$143.47**

8.4 × 2.03/12 lb/in × $38.72/lb
saw/drill: (0.1667 + 0.0167·n) h × $100/h
optional face both ends: +$145.00 → $288.47

WEIGHT · 6061, 0.0975 lb/in³

Stock 1 blank, 8.20 in	1.380
4 × Ø1.00 web holes	−0.046
12 top-flange holes	−0.146
16 bottom-flange holes	−0.194
Finished beam (lb)	**0.993**

volumes from the Abaqus CAD model
margin to the 1.00 lb limit: 0.007 lb

HOLE SCHEDULE

Tag	Size	Qty	Location (from CL, web centreline)	
A	Ø1.00	4	web, z ±1.00 (y +0.375), ±3.00 (y −0.375)	required
B	Ø0.78125 (25/32)	12	top flange, x ±0.7875, z ±1.30, ±2.18, ±3.06	drill thru
C	Ø0.78125 (25/32)	16	bottom flange, x ±0.7875, z ±0.44, ±1.32, ±2.20, ±3.08	drill thru

TITLE	REV	MATERIAL
Stock 1 Drilled I-Beam, Final Design	**v2**	6061-T6 Al
PROJECT	**DATE**	**UNITS / TOL**
EMCH 308 · USC Team 6 / MTC	2026-10-26	in · drill ±0.02

Notes: deburr all holes · no milling required · holes drilled through one flange only (do not cut into web or fillets) · source: abaqus/python/final/Stock1_Drilled_v2.py